Mensik_Exam

Kristina Mensik

2026-10-19

Question 6

```
data <- read.csv("/Users/kom11/Documents/GitHub/regression-study/Experiments/green_study.csv")

data <- na.omit(data)

getwd()
```

```
## [1] "/Users/kom11/Documents/GitHub/regression-study/Experiments/Section"
```

6.1

2716/4370 voted in 98, or 62% 1998 out of 7086 or 28% of the sample voted both years.

```
names(data)
```

```
## [1] "ID"        "Vote98"    "Contact"   "Contacted" "Age"       "Party"
## [7] "Vote96"
```

```
table(data$Vote98)
```

```
##
##    0    1
## 4370 2716
```

```
nrow(data)
```

```
## [1] 7086
```

```
2716/4370
```

```
## [1] 0.6215103
```

```
table(data$Vote98, data$Vote96)
```

```
##
##        0    1
##   0 3126 1244
##   1  718 1998
```

```
1998/7086
```

```
## [1] 0.2819644
```

6.2

non compliance? contact and contacted

```
table(data$Contact)
```

```
##
##    0    1
## 5645 1441
```

```
table(data$Contact, data$Contacted)
```

```
##
##        0    1
##   0 5645    0
##   1 1046  395
```

Yes there was noncompliance: the above table shows that of the 1441 assigned to treatment, only 395 were contacted or treated. There are compliers and never takers in the study. No people assigned to control got treated. The control group may have never takers but we don't know the difference between them and compliers.

6.3

```
ITT_Y <- mean(data$Vote98[data$Contact == 1]) -
       mean(data$Vote98[data$Contact == 0])

print(ITT_Y)
```

```
## [1] 0.03891919
```

```
ITT_D <- mean(data$Contacted[data$Contact == 1]) -
         mean(data$Contacted[data$Contact == 0])

print(ITT_D)
```

```
## [1] 0.2741152
```

```r
CACE <- ITT_Y / ITT_D
print(CACE)
```

```
## [1] 0.1419811
```

```r
model_cace <- iv_robust(Vote98 ~ Contacted | Contact, data = data)
```

For compliers, the average treatment affect of being canvaassed was 0.14

###6.4

```r
#Step 1: calculate the observed F statistic

model_obs <- lm_robust(
  Contact ~ Age + Party + Vote96,
  data = data
)

F_obs <- model_obs$fstatistic[1]

# The F stat would describe how strongly the three pretreatment covariates jointly predict the treatmen

# Need to compare it to the distribution of F statistics generated by randomization.

# Step 2: delcare the randomization procedure (the procedure for assigning N and m needs to produce the

ra <- declare_ra(
  N = nrow(data), # total num observations
  m = sum(data$Contact == 1) # num assigned to treatment
)

# Step 3: Initialize a vector that will eventualyl hold F stats

B <- 1000 # number of simulated random draws reducing so I can knit (running at 10k)
F_sim <- numeric(B)

# Step 4: Conduct RI (only the assignment vector changes across iterations)

for (i in 1:B) {

  data$Z_sim <- conduct_ra(ra)

  model_sim <- lm_robust(
    Z_sim ~ Age + Party + Vote96,
    data = data
  )

  F_sim[i] <- model_sim$fstatistic[1]
}

# Step 5: p value

p_value <- mean(F_sim >= F_obs)
```

```
# interpretation: under the specified RA procedure, approx (p_value)% of hypothetical assignments produ

#
```

Question 5

5 a

Yes – This might change the assignments that can occur. I'd need to change how I estimate the CACE because this would require using inverse probability weights.

5 b

```
library(broom)


restricted_ra <- function(data, m) {

  while (TRUE) {

    # Generate hypothetical treatment assignment
    Z_temp <- complete_ra(N = nrow(data), m = m)

    # Regress hypothetical assignment on covariates
    model <- lm_robust(
      Z_temp ~ Age + Party + Vote96,
      data = data
    )

    # Extract F-statistic and degrees of freedom
    f <- model$fstatistic

    # Calculate upper-tail F-test p-value
    p_balance <- pf(
      q = f[1],
      df1 = f[2],
      df2 = f[3],
      lower.tail = FALSE
    )

    # ACCEPT assignment only if p > 0.10
    if (p_balance > 0.10) {
      return(Z_temp)
    }
  }
}

# STEP 2: Generate 1k acceptable assignments

set.seed(6081709)
```

```
B <- 1000
m <- sum(data$Contact == 1)

sim_matrix <- replicate(
  B,
  restricted_ra(data = data, m = m)
)

# Each  is one acceptable assignment vector
dim(sim_matrix)

# STEP 3: declare

ra_restricted <- declare_ra(
  permutation_matrix = sim_matrix
)

print(ra_restricted)

# STEP 4: Get inverse-probability weights
# Using the OBSERVED assignment Z

data$weights <- 1 /
  obtain_condition_probabilities(
    ra_restricted,
    data$Contact
  )

# STEP 5: Estimate weighted CACE with IV regression

model_cace <- iv_robust(
  Vote98 ~ Contacted | Contact,
  weights = weights,
  data = data
)

# Estimate SE, p-value, CI
print(tidy(model_cace))
```